

Theoretical Proof that Auto-regressive Language Models Collapse when Real-world Data is a Finite Set

Lecheng Wang^{1,2}, Xianjie Shi^{1,2}, Ge Li^{1,2*}, Jia Li^{♂1,2}, Xuanming Zhang³,
Yihong Dong^{1,2}, Wenpin Jiao^{1,2}, Hong Mei^{1,2}

¹Key Laboratory of High Confidence Software Technologies (Peking University), Ministry of Education

²School of Computer Science, Peking University, Beijing, China

³University of Wisconsin-Madison

wanglecheng@stu.pku.edu.cn, lige@pku.edu.cn

Abstract

Auto-regressive language models (LMs) have been widely used to generate data in data-scarce domains to train new LMs, compensating for the scarcity of real-world data. Previous work experimentally found that LMs collapse when trained on recursively generated data. This paper presents a theoretical proof: once a corpus (such as a subset of the World Wide Web) begins to incorporate generated data and no new real-world data is added to the corpus, then no matter how small the amount of data each LM generates and contributes to the corpus, LM collapse is inevitable after sufficient time. This finding suggests that attempts to mitigate collapse by limiting the *quantity* of synthetic data in the corpus are fundamentally insufficient. Instead, avoiding collapse hinges on ensuring the *quality* of synthetic data.¹

1 Introduction

Auto-regressive language models (LMs)² like ChatGPT (OpenAI et al., 2024) and Llama (Dubey et al., 2024) are widely used to generate text on the World Wide Web, such as articles and computer source code. Meanwhile, practitioners collect large amounts of text from the Web to train the next generations of LMs. Thus, the collected training corpus will inevitably include text generated by existing LMs. However, it is unclear whether the generated text in the training corpus has an inevitable negative impact on LMs (Chen et al., 2024). Clarifying this issue is crucial for model trainers who rely on scraping a large amount of unvetted, potentially synthetic data from the Web to train LMs. It also critically applies to trainers using synthetic data from LMs to compensate for the scarcity of real-world data sources. This includes scenarios like using chain-of-thought data when training instruct models (Wang et al., 2023; Xu et al., 2024), or leveraging synthetic data in fields such as physics or computational biology when training corresponding large models (CAMEL-AI.org, 2025).

A pioneer study (Shumailov et al., 2024) warned that AI models, including LMs, can *collapse* when trained on recursively generated data. Studies related to model collapse (Taori & Hashimoto, 2023; Alemohammad et al., 2024; Bertrand et al., 2024; Gerstgrasser et al., 2024; Kazdan et al., 2024) usually train the first-generation model with initial data, mix the data generated by the $(n - 1)$ -th generation model with previous data in different proportions to train the n -th generation model, and observe whether the evaluation performance of the model deteriorates, *i.e.*, the model *collapses*. Although these studies revealed potentially adverse effects of generated data on many LMs, *e.g.*,

*Corresponding author

¹The source code is available in the online data warehouse: <https://github.com/wanglc02/generated-data>

²In this paper, unless otherwise specified, LM refers to an *auto-regressive* language model.

OPT-125M (Zhang et al., 2022), GPT, and Llama, it does not necessarily follow that all LMs will collapse if trained in the above-mentioned way. Therefore, a theoretical proof of LM collapse is essential.

Previous theoretical works proved that Gaussian model (Shumailov et al., 2024), linear regression model (Dohmatob et al., 2024a; Gerstgrasser et al., 2024), simplified (non-autoregressive) LM (Seddik et al., 2024), and toy bigram LLMs (large language models) (Dohmatob et al., 2024c) can collapse. Dohmatob et al. (2024b) showed that even the smallest fraction of synthetic data (*e.g.*, as little as 1% of the total training dataset) can still lead to the collapse of linear model and random projection model. However, to our knowledge, no one has proven that auto-regressive LMs will collapse. We believe that simply drawing analogies between LMs and other generative models cannot prove that LMs also conform to these collapse laws.

Recognizing this research gap, we present the first theoretical proof of collapse in auto-regressive LMs, demonstrating such collapse in data-scarce domains—specifically, when the real-world data constitutes a finite set. Our motivation is to hope that the theoretical analysis approach we have discovered can assist researchers in using theoretical tools to further explore the dynamics of LM collapse in domains with more complex data sources. In addition, we prove that LM collapse occurs regardless of the rate at which generated data infiltrates the initial real-world dataset in our scenario. This reveals that preventing LM collapse seems to rely more on filtering out low-quality data before training than on restricting generated data from entering the training corpus.

2 Proof: language model collapse

Kazdan et al. (2024) empirically observed that auto-regressive LMs collapse when recursively trained under the condition that real-world data constitutes a finite set. This scenario encompasses two data paradigms discussed in their paper: *Replace* and *Accumulate-Subsample*.

In the *Accumulate-Subsample* paradigm, a corpus progressively incorporates generated data, from which training samples are drawn for each LM. Our core contribution is a proof of LM collapse in this paradigm.

The *Replace* paradigm is a classic paradigm first proposed by Shumailov et al. (2024). In this paradigm, each LM can only access the data generated by the previous LM during training. Notably, the collapse of LMs in this paradigm has remained unproven until now.

As Kazdan et al. (2024) noted, *Replace* is an unrealistic model of reality: in practice, we do not discard real-world data after training our first LM. By contrast, *Accumulate-Subsample* reflects data accumulation (analogous to real-world Web data growth), where LMs are trained under a fixed computational budget—a scenario aligned with the resource constraints of individual practitioners. Given the similarity of proof strategies for both paradigms, we address them jointly. Our proof proceeds through the following steps:

- Section 2.1 formally defines the model of *LM collapse* and establishes the foundational assumptions underpinning our Main Theorem.
- Section 2.2 formulates the Main Theorem in a self-contained manner.
- Sections 2.3 to 2.6 present the theorem’s proof through four phases:
 - Section 2.3 quantifies non-negative errors between the output distribution of generation 1 and the distribution of the initial corpus.
 - Section 2.4 presents the errors from generation to generation.
 - Section 2.5 derives a closed-form expression for the output distribution of generation n in terms of the distribution of the initial corpus and the errors introduced in each generation.
 - Section 2.6 demonstrates that these errors form positive series as $n \rightarrow \infty$, causing the output distribution of generation n to approach arbitrarily closely a term composed solely of the errors introduced in each generation.
- Finally, Section 2.7 points out that the amount of generated data incorporated into the corpus in each generation does not affect the proof process. No matter how restricted the amount of data generated by each LM that flows into the corpus is, LM collapse is inevitable.

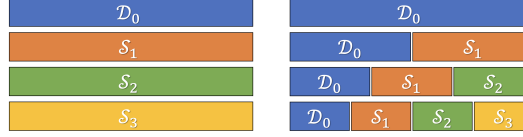


Figure 1: **Composition of the training set for each generation of language models (LMs) under different data paradigms.** On the left is *Replace*, and on the right is a special case of *Accumulate-Subsample* when $k = 1$. From top to bottom, each rectangle represents the composition of the training set of the 1st, 2nd, 3rd, and 4th generation LMs, respectively. $\mathcal{D}_0$, $\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$ represent the initial corpus, the data generated by the 1st-generation LM, by the 2nd-generation LM, and by the 3rd-generation LM, respectively. The original figure is Figure 2 in the paper by Briesch et al. (2024).

2.1 Assumptions

Before presenting our Main Theorem, we make two assumptions. First, the two data paradigms used to train each LM involved in our theorem are defined as described in Section 2.1.1 and Figure 1. Second, the meanings of all notations follow the specifications in Section 2.1.2.

2.1.1 Definitions

Replace The first LM (generation 1) is trained on an initial corpus. The n -th LM (generation n) is trained on the data generated by the $(n - 1)$ -th LM for each $n \geq 2$.

Accumulate-Subsample The first-generation LM is trained on an initial corpus. For each subsequent generation $n \geq 2$, the n -th LM is trained on the same total volume of data as generation 1. Of this data, a proportion $\frac{1}{1+(n-1)k}$ is sourced from the initial corpus. Meanwhile, each of the preceding $n - 1$ LMs (from generation 1 to $n - 1$) contributes a proportion $\frac{k}{1+(n-1)k}$ of the data, where $k > 0$ is a real-valued parameter representing the rate at which generated data is integrated into the corpus. This is because we assume that the data *accumulate* in the following way: The data generated by generation 1 is added to the initial corpus, and the amount of data added to the corpus is k times the initial data amount. If k is equal to 1, then for each generation, the amount of generated data added to the corpus is as large as the initial data amount. This special case is the *balanced* data cycle among the four data cycles defined by Briesch et al. (2024). Subsequently, a subset as large as the initial corpus is *subsampled*³ from the mixed corpus as the training set for generation 2. The data generated by generation 2 is added to the corpus, increasing the amount of data in the corpus by an amount equivalent to k times the amount of data in the initial corpus. The training set for generation 3 is also *subsampled* from the mixed corpus, and so on. In this way, the amount of data in the corpus grows, while the amount of training data for each generation of LMs is constant.

2.1.2 Notations

- V is a finite set of tokens. Tokens are the basic units in LMs, often being words or meaningful subwords. Common tokens are collected as a vocabulary V .
- $\mathbf{x} \in V^*$ denotes a sequence of tokens.
- Vector $(p(v_1|\mathbf{x}), p(v_2|\mathbf{x}), \dots)$ is a probability distribution on the vocabulary V . It denotes the ground-truth distribution of the initial corpus. $\forall v_i \in V$ $p(v_i|\mathbf{x}) \triangleq \frac{N_{\mathbf{x};v_i}}{N_{\mathbf{x}}}$, where $N_{\mathbf{x}}$ denotes the number of occurrences of $\mathbf{x}$, $N_{\mathbf{x};v_i}$ denotes the number of occurrences of the concatenation of $\mathbf{x}$ and v_i in the corpus, subject to $N_{\mathbf{x}} = \sum_{v_i \in V} N_{\mathbf{x};v_i}$.
- An LM takes an $\mathbf{x}$ (often called ‘context’ or ‘prompt’) as input.
- The LM at generation n outputs a vector $(\hat{p}_n(v_1|\mathbf{x}), \hat{p}_n(v_2|\mathbf{x}), \dots)$, a probability distribution representing its prediction of the next token v_i on the vocabulary V . **Assume that:**

³Assume that the initial corpus is so large that the sampling error in the sampling process can be ignored. We will prove that even if the sampling error is ignored, the LMs will still collapse.